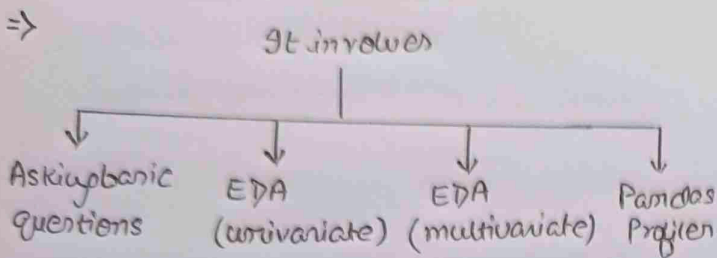


EDA

Understanding Your Data :



⇒ Some basic question about data shall be clear, such as :

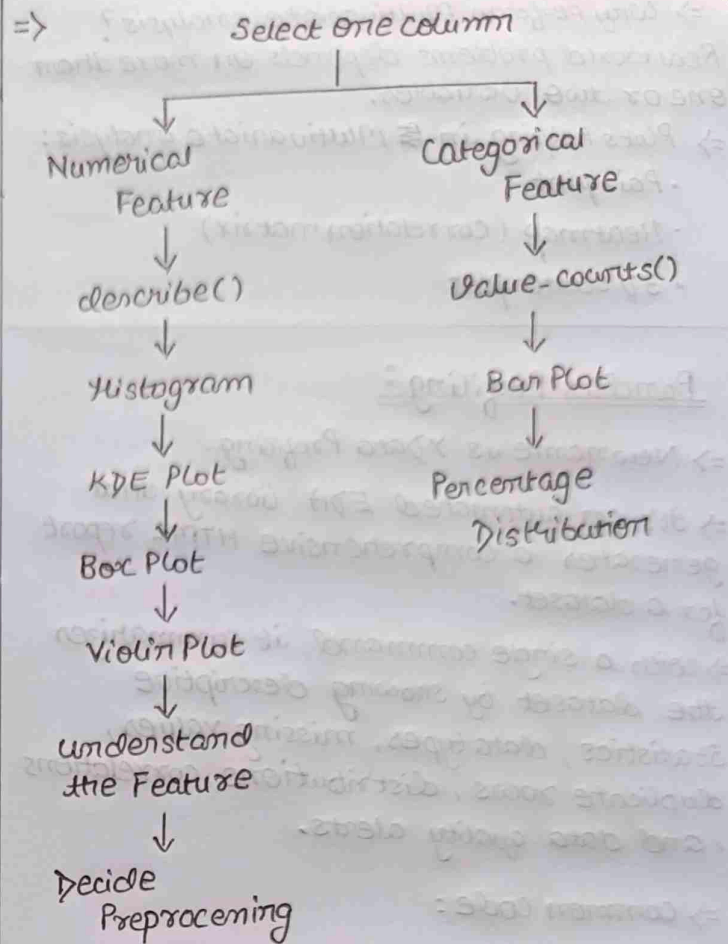
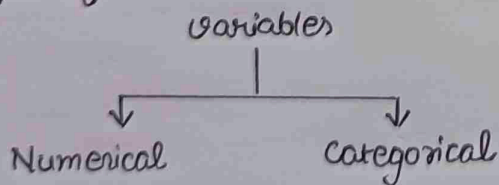
- ① How big is the data?
- ② How does the data look?
- ③ What is the data type of each column?
- ④ Are there any missing values?
- ⑤ How does the data look mathematically?
- ⑥ Are there duplicate values?
- ⑦ How are the columns related?

EDA - Univariate Analysis :

⇒ What is EDA? Exploratory Data Analysis helps us explore the data visually and statistically before feature engineering or model building.

⇒ What is Univariate Analysis? Study one column at a time without considering its relationship with other columns.

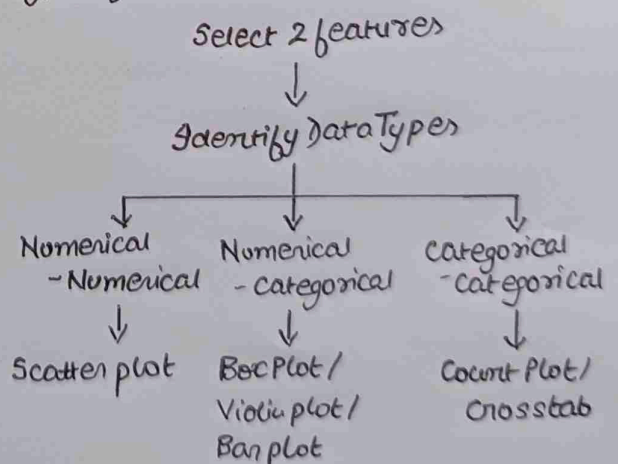
⇒ Types of variables in univariate analysis:



EDA - Bivariate & Multivariate Analysis :

⇒ univariate analysis → one variable
Bivariate analysis → Two variables
Multivariate analysis → three or more variables

⇒ Types of Bivariate analysis :



=> Why perform Multivariate analysis?
 Real world problems depends on more than one or two variables.

=> Plots helping in Multivariate analysis:

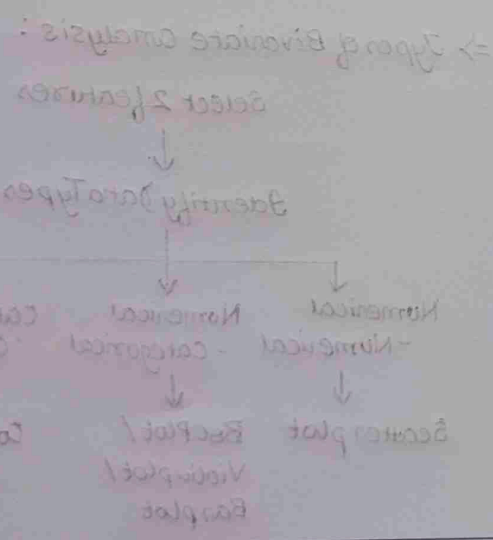
- Pair plot
- Heatmap (Correlation matrix)
- 3D scatter plot

Pandas Profiling:

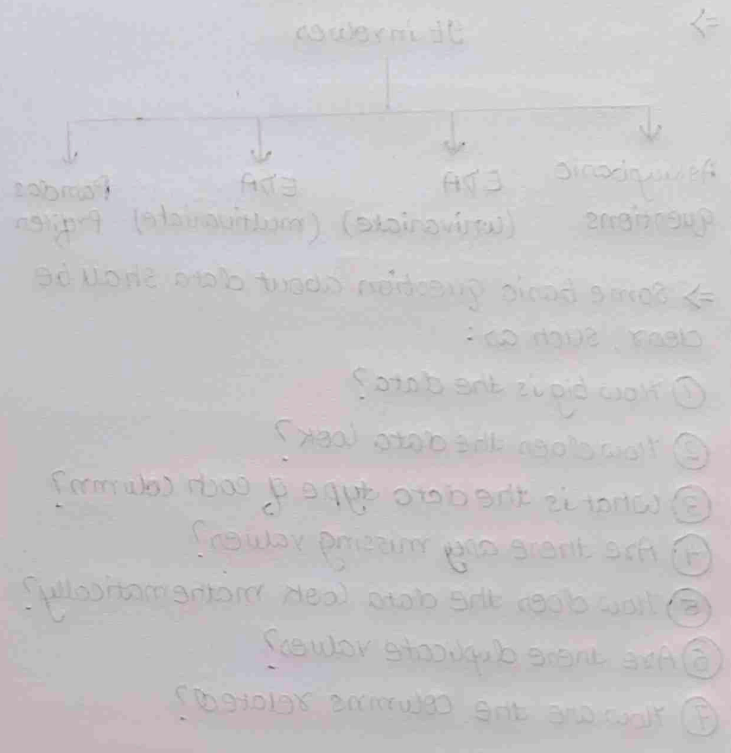
=> New name is YData Profiling.
 => It's an automated EDA library that generates a comprehensive HTML report for a dataset.
 => With a single command, it summarizes the dataset by showing descriptive statistics, data types, missing values, duplicate rows, distributions, correlations, and data quality alerts.

=> Common code:

```
import pandas as pd
from ydata-profiling import ProfileReport
df = pd.read_csv("data.csv")
profile = ProfileReport(df, title="EDA Report")
profile.to_file("report.html")
```



Understanding your data:



EDA - Univariate analysis:

=> What is EDA? Exploratory Data Analysis helps us explore the data visually and statistically before feature engineering or model building.
 => What is Univariate analysis? Study one column at a time without considering its relationship with other columns.

